

RAHAGEER SAADMAN ISLAM

AI/ML Engineer | Agentic AI | LangChain | LLMs | RAG | MCP | FastAPI | Python

rsi.rahageer@gmail.com | linkedin.com/in/rahageer-rsi | github.com/Rahageer90 | Dhaka, Bangladesh

WORK EXPERIENCE

AI/ML Engineer (MLOps) — Deep Data Labs / Cloudscockpit

Dhaka, Bangladesh | January 2026 – Present

- Designed and shipped **11 MCP-based AI tools** tailored for an **NGO use case** into the Cloudscockpit genai-core platform — including neural search, verify information, research query, get news sources, and computer vision image analysis tools.
- Built and shipped a **Marketing MCP server** comprising four production tools: *monitor_trends* for real-time trend tracking, *competitor_analysis* for automated competitive intelligence, *post_suggestion* for AI-driven content recommendations, and *time_tools* for scheduling and temporal context — enabling agents to execute end-to-end marketing workflows autonomously.
- Extended the **Document Generator MCP server** with a *build_pptx* tool that allows AI agents to programmatically generate PowerPoint presentations with embedded Seaborn charts; engineered **4 distinct design templates** (corporate, minimal, data-heavy, and creative) to support varied presentation contexts.
- Organized the file manager for the **ActionBoard** product, architecting dedicated storage directories (e.g., *saved_pdfs*, *saved_docx*, *saved_images*) to systematically categorize and persist agent-generated files per board.
- Shipped the **Alops Maturity Metrics** feature for ActionBoard to evaluate user AI execution proficiency, calculated by tracking completed guided agent cycles, tool call volumes, and contextual skills added per board.
- Collaborated in a multi-developer environment alongside 4 contributors, deploying reliable data retrieval endpoints with structural fallback mechanisms.

Stack: Python, FastAPI, LangChain, MCP, Vector Databases, Google Gemini API, NewsAPI, Serper API, Python-pptx, Seaborn, Docker

SKILLS

AI/ML Core:	Large Language Models (LLMs), NLP, Generative AI, Computer Vision, Explainable AI (XAI), Anomaly Detection, Classification
Agentic AI & RAG:	LangChain, LangGraph, Model Context Protocol (MCP), ReAct Agents, RAG, Vector Embeddings, Semantic Search
Web & Data:	PHP, PostgreSQL, MySQL, MVC Architecture, REST API Design, HTML/CSS, Tailwind CSS
ML Libraries:	Scikit-learn, Pandas, NumPy, Jupyter Notebook, feature engineering, cross-validation, ensemble methods
APIs & LLM Providers:	Google Gemini API, Serper API, NewsAPI, Tavily API, OpenAI-compatible endpoints, FastAPI
MLOps, Cloud & Infra:	Python (98.5%), Docker, Kubernetes, Basic AWS (ECS, S3, ElastiCache, RDS, Lambda), Redis, Netlify, Aiven, CI/CD for ML

PROJECTS

Fact-Check Agent — Autonomous Verification System

- Architected a full-stack agentic AI system for autonomous claim verification, orchestrating multiple LLM providers (Google Gemini API) with real-time data retrieval.
- Engineered 2 MCP-based tools: a Serper API-based *web_search* and a News API-based *news_search*, each with fallback mechanisms for production reliability.
- Built a ReAct agent loop with LangChain for multi-step reasoning: query decomposition, evidence retrieval, cross-source reconciliation, and confidence-scored verdict generation.
- Deployed FastAPI REST backend and React frontend; demonstrates enterprise-scale agentic system architecture.

Stack: Python, FastAPI, LangChain, MCP, Google Gemini API, Serper API, NewsAPI, React

Undergraduate Thesis: Hybrid Learning-Based Intrusion Detection Framework with LIME-Driven Interpretability

- Developed ensemble ML models (classification + anomaly detection) to identify malicious network traffic patterns across multiple attack categories.

- Implemented Explainable AI (XAI) techniques — feature importance analysis and LIME-based model interpretability — ensuring transparent decision-making critical for security applications.
- Delivered a complete ML pipeline: data preprocessing, feature engineering, model training, cross-validation, hyperparameter tuning, and performance benchmarking.

Stack: Python, Scikit-learn, XAI / LIME, Pandas, NumPy, Jupyter Notebook

Healthcare ML — Diabetes Risk Prediction System

- Engineered a supervised ML model for diabetes risk assessment: full pipeline covering data cleaning, feature engineering, model selection, and statistical validation.

Stack: Python, Scikit-learn, Pandas, NumPy

EDUCATION

B.Sc. in Computer Science & Engineering

BRAC University, Dhaka, Bangladesh

Oct 2021 – Jan 2026

Higher Secondary Certificate, Science

Adamjee Cantonment College, Dhaka, Bangladesh

Jul 2018 – Apr 2020

CERTIFICATIONS

- Python for Data Science, AI & Development — IBM
- Data Analysis with Python — IBM
- Introduction to Generative AI — Google
- Applied Data Science Capstone — Coursera